

Question

The carbon dioxide gas-sensing electrode consists of a gas-permeable membrane, a $0.010 \text{ mol} \cdot \text{L}^{-1}$ NaHCO_3 solution, a pH electrode, and an external reference electrode. The external reference electrode is a standard calomel electrode with $\varphi(\text{Cl}^- | \text{Hg}_2\text{Cl}_2 | \text{Hg}) = 0.2801 \text{ V}$. Calculate the electrode potential of the galvanic cell when the mass fraction of carbon dioxide in the atmosphere is 611 ppm by mass . Given: Henry's law constant for CO_2 is $K_{\text{H}} = 3.0 \times 10^3 \text{ kPa} \cdot \text{L} \cdot \text{mol}^{-1}$, the equilibrium constant for $\text{CO}_2 + \text{H}_2\text{O} = \text{H}_2\text{CO}_3$ is $K = 1.70 \times 10^{-3}$, and the acid dissociation constants of carbonic acid are $\text{p}K_{\text{a}1,2} = 6.35, 10.33$.

A. 0.254 V

B. 0.397 V

C. 0.543 V

D. 0.689 V

E. 0.731 V

F. 0.826 V

G. 1.031 V

H. 1.102 V

Answer

Correct Answer: **F**

Detailed Explanation

From $\omega_{\text{CO}_2} = 611$ ppm by mass, i.e., $\omega_{\text{CO}_2} = 6.11 \times 10^{-4}$, and the molar mass and standard atmospheric pressure data $M_{\text{CO}_2} = 44 \text{ g}\cdot\text{mol}^{-1}$, $\bar{M}_{\text{air}} = 29 \text{ g}\cdot\text{mol}^{-1}$, $1 \text{ atm} = 101.325 \text{ kPa}$, we calculate: $p_{\text{CO}_2} = \omega_{\text{CO}_2} \times \frac{\bar{M}_{\text{air}}}{M_{\text{CO}_2}} \times 1 \text{ atm} = 0.0403 \text{ kPa}$.

CHECKPOINT

1 PTS

$$p_{\text{CO}_2} = 0.0403 \text{ kPa}$$

From Henry's law, the concentration of CO_2 at dissolution equilibrium in the solution is $[\text{CO}_2] = \frac{p_{\text{CO}_2}}{K_{\text{H}}} = 1.36 \times 10^{-5} \text{ mol}\cdot\text{L}^{-1}$.

CHECKPOINT

1 PTS

$$[\text{CO}_2] = 1.36 \times 10^{-5} \text{ mol}\cdot\text{L}^{-1}$$

In this system, $[\text{HCO}_3^-]$ is much larger than $[\text{H}_2\text{CO}_3]$, so it can be approximately considered that $[\text{HCO}_3^-] = 0.010 \text{ mol}\cdot\text{L}^{-1}$ remains unchanged.

CHECKPOINT

1 PTS

$$[\text{HCO}_3^-] = 0.010 \text{ mol}\cdot\text{L}^{-1} \text{ remains unchanged}$$

From ionization equilibrium, $[\text{H}^+] = \frac{K_{\text{a1}}[\text{CO}_2]}{[\text{HCO}_3^-]} = \frac{10^{-6.35} \times 1.36 \times 10^{-5}}{0.010} = 6.07 \times 10^{-10} \text{ mol}\cdot\text{L}^{-1}$, i.e., $\text{pH} = 9.22$.

CHECKPOINT

1 PTS

$$\text{pH} = 9.22$$

The standard calomel electrode is the anode, and the cathode is where water is reduced to produce hydrogen. $\varphi_{\text{anode}} = 0.2801 \text{ V}$, $\varphi_{\text{cathode}} = 0.0592 \lg [\text{H}^+] = -0.546 \text{ V}$, so $E = (0.2801 + 0.546) = 0.826 \text{ V}$.

CHECKPOINT

1 PTS

$$E = 0.826 \text{ V}$$